

Rounds 1 & 2

- Round1
- Round2

Round 1

Correct Answer

The screenshot shows a web browser window displaying a Canvas LMS quiz. The address bar shows the URL: <https://canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693005>. The page title is "Quiz: Basic level of the exam (13)". A green notification box at the top says: "You started this quiz near when it was due, so you won't have the full amount of time to take the quiz." The main content area is titled "Quiz Instructions" and states: "Only one question is shown at a time." Below this, "Question 15" is displayed, worth "1 pts". The question text is: "If a real number set has minimum, then it is bounded above." There are two radio button options: "True" and "False". The "False" option is selected. A blue handwritten note next to the question says: "false , tu minusisken ari bounded above". To the right of the question, a list of questions is shown, with "Question 2" through "Question 8" marked as completed. Below the list, it says "Time Running: Hide" and "Attempt due: Jun 13 at 3:50pm 30 Minutes, 5 Seconds". At the bottom of the question box, there is a "Previous" button and a "Submit Quiz" button. The bottom of the browser window shows a Windows taskbar with the date and time: "3:19 PM 6/13/2023".

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693001

Only one question is shown at a time.

Question 9 1 pts

Let $\emptyset \neq H \subseteq \mathbb{R}$. The set H is open set if and only if

A) Any point in it is interior point
 B) It has at least one interior point
 C) It has infinitely many interior points
 D) It has no boundary point

☐ B)
☐ D)
☒ A)
☐ C)

Handwritten notes: *correct D*, *open = no boundary points*, *open (3;5)*

Time Running: Hide
 Attempt due: Jun 13 at 3:50pm
 31 Minutes, 9 Seconds

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Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693145

Only one question is shown at a time.

Question 2 1 pts

Let $a_n, b_n, c_n \in \mathbb{R}$ $n \in \mathbb{N}$.

A) If $\lim(b_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(a_n) = A$
 B) If $\forall n \in \mathbb{N} : a_n \leq b_n \leq c_n$ and $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$
 C) If $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$
 D) If $\lim(a_n) = \lim(b_n) = A \in \mathbb{R}$, then $\lim(c_n) = A$

☐ C)
☒ B)
☐ A)
☐ D)

Handwritten notes: *same limit*, *answer B*, *sandwich theorem*

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 Attempt due: Jun 13 at 3:50pm
 43 Minutes, 23 Seconds

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Quiz: Basic level of the exam (13)

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Question 3 1 pts

4. State the Cauchy convergence test for sequences.

A) A real number sequence is a Cauchy-sequence if and only if it is convergent.
B) A sequence $(a_n) : \mathbb{N} \rightarrow \mathbb{R}$ is convergent if and only if $\lim(\sqrt[n]{|a_n|}) < 1$.
C) A real number sequence is a Cauchy-sequence if and only if it has limit.
D) A real number sequence (a_n) is convergent if and only if

$$\exists N \in \mathbb{N} \forall \varepsilon > 0 \forall m, n \geq N : |a_n - a_m| < \varepsilon$$

☒ D)
☐ A)
☐ C)
☐ B)

has limit $\rightarrow$ converges

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3:08 PM 6/13/2023

Handwritten notes:
Caish $\rightarrow$ Con
Con $\nrightarrow$ Ca
answer D
Given sequence
any two $\rightarrow$
their dif $\rightarrow -\infty$

Quiz: Basic level of the exam (13)

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692993

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Only one question is shown at a time.

Question 4 1 pts

Let (a_n) be a zero sequence, (b_n) be a convergent sequence, $\lim(b_n) = 2$. Then

☐
☒ $\lim(a_n b_n) = 0$
☐ $\lim(a_n b_n) = -1$
☐ $\lim(a_n b_n) = 1$

answer B

all convergent = bounded

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3:09 PM 6/13/2023

Handwritten notes:
 $\lim = 0$
any bounded sequence $= 0$
multiplied by

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692997

Question 6 1 pts

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
38 Minutes, 41 Seconds

Let $q \in \mathbb{K}$, $|q| < 1$. Then

A) $\sum_{n=0}^{\infty} q^n = \frac{q}{1-q}$

B) $\sum_{n=1}^{\infty} q^n = \frac{1}{1-q}$

C) $\sum_{n=0}^{\infty} q^n = \frac{1}{(1-q)^2}$

D) $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$ **answer D**

$|q| < 1$
 $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$
 $\sum_{n=1}^{\infty} q^n = \frac{q}{1-q}$

Geometric sequence sum

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Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693147

Quiz You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Only one question left

Question 7 1 pts

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
31 Minutes, 35 Seconds

4. Which of the following statements is true?

A) If $\sum a_n$ is convergent, then $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$

B) If a series is convergent then it is absolutely convergent.

C) If $\lim(a_n) = 0$, then $\sum a_n$ is convergent.

D) If $\sum a_n$ is convergent, then $\lim(a_n) = 0$.

answer - D

$\sum a_n$
 $\sum_{n=0}^{\infty} \frac{1}{n}$
 a_n
 $|a_n|$
both

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Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693136

Only one question is shown at a time. You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Question 8 1 pts

Choose the correct answer for

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{6x} = ?$$

☐ 0
☒ $\frac{1}{2}$
☐ LaTeX: 1
☐ 2

$\lim_{x \rightarrow 0} \frac{\sin 3x}{6x} = \frac{1}{2} \cdot \frac{\sin 3x}{3x} = \frac{1}{2}$

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Time Running: Hide
Attempt due: Jun 13 at 3:50pm
31 Minutes, 26 Seconds

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Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693140

Only one question is shown at a time. You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Quiz Instructions

Question 11 1 pts

15. If a function of type $\mathbb{R} \rightarrow \mathbb{R}$ is continuous, then it has minimum and it has maximum.

☐ True
☒ False

$\int U$

doesn't have both ans → false

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Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 45 Seconds

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Quiz: Basic level of the exam (13 June 2023) (15 questions)

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Quiz Instructions

Only one question is shown at a time.

Question 12 1 pts

9. If a nonempty real number set is bounded below, then it has minimum.

☐ True

☒ False *answer: false*

3 *no minimum*

Not saved Submit Quiz

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 37 Seconds

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2022/23/2 IP-18fAN1E 1

2022/23/2

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Grades

Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 13 1 pts

11. If the limit of a number sequence is $-\infty$, then it is not bounded below.

☒ True *answer: true*

☐ False

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 28 Seconds

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$$|a_n| \leq M$$

$$M \in \mathbb{R}$$

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693139

Quiz Instructions

Only one ✓ You started this quiz near when it was due, so you won't have the full amount of time to take the quiz. X

Question 14 1 pts

If the terms of an infinite series converge to zero, then the series is convergent.

☐ False *answer: FALSE*

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✓ Question 5

✓ Question 6

✓ Question 7

✓ Question 8

Time Running: [Hide](#)

Attempt due: Jun 13 at 3:50pm

30 Minutes, 15 Seconds

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In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.



Question 5

1 pts

Let $q \in \mathbb{K}$, $|q| < 1$. Then

A) $\sum_{n=0}^{\infty} q^n = \frac{q}{1-q}$

B) $\sum_{n=1}^{\infty} q^n = \frac{1}{1-q}$

C) $\sum_{n=0}^{\infty} q^n = \frac{1}{(1-q)^2}$

D) $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$

answer
D

☐ A)

☐ B)

☐ C)

☒ D)



Only one question is shown at a time.

monotonically decreasing + convergent to 0

1 pts

$$\frac{(-1)^h}{n^2 + 1}$$

A) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

$$B) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$$
$$c) \sum_{n=1}^{\infty} \frac{-1}{2n}$$
$$D) \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n} \quad \frac{1}{n}$$

☒ A)

☐ B)

☐ C)

☐ D)
$$a_n$$

$|a_n|$
absolutely

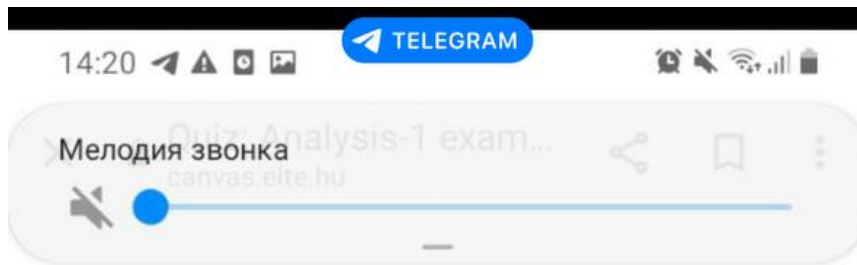
$\left(\frac{1}{n}\right) \rightarrow \text{divergent}$

diversent
absolute
Lend- und
Kontoposen

Next >

Submit

<



the question is shown at a time.

Question 7**1 pts**

Provided that it has at least two elements, the convergence set of the real power series $\sum_{n=0}^{\infty} a_n \cdot (x - x_0)^n$ is:

- ☐ an open interval centered at x_0
- ☒ an interval centered at x_0
- ☐ a closed interval centered at x_0
- ☐ not always an interval

answer: B
can be both
closed or open

only one
point

interval

open $(x_0 - r; x_0 + r)$
close $[x_0 - r; x_0 + r]$
 $[x_0 - r; x_0 + r)$
 $(x_0 - r; x_0 + r]$

◀ Previous

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Not saved Submit Quiz



14:20

TELEGRAM

Quiz: Analysis-1 exam...
canvas.elte.hu

In questions 1-10 mark the single correct answer

In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.

Question 8

1 pts

Choose the correct answer for

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{6x} = ?$$

☐ 1

☐ 2

☒ $\frac{1}{2}$ *answer*

☐ 0

◀ Previous

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Not saved

Submit Quiz

one question is shown at a time.



Question 9

6. Define the sin function.

$$A) \quad \sin(x) := \sum_{n=1}^{+\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$$

$$B) \quad \sin(x) := \sum_{n=0}^{+\infty} \frac{(-x)^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$$

$$C) \quad \sin(x) := \sum_{n=0}^{+\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$$

$$D) \quad \sin(x) := \sum_{n=0}^{+\infty} \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$$

☐ A)

☐ B)

☒ C)

☐ D)

◀ Previous

D

Only one question is shown at a time.



Question 10

1 pts

Let $\emptyset \neq H \subseteq \mathbb{R}$. The set H is open set if and only if

- A) Any point in it is interior point
- B) It has at least one interior point
- C) It has infinitely many interior points
- ☒ D) It has no boundary point

— open $\rightarrow$ no boundary

closed
 $\downarrow$
has boundary
 $\uparrow$
All points

☒ A)

☐ B)

☐ C)

☐ D)

◀ Previous

Next ▶

Not saved

Submit

Questions



one question is shown at a time.

Question 11

9. If a real number set has minimum, then it is bounded above.

☐ True

☒ False

◀ Previous

Not saved

Questions 1-10 mark the single correct answer.

Questions 11-15 mark that the statement is true or false.

One question is shown at a time.

Question 12

12. If $(a_n + b_n)$ is a zero sequence, then either (a_n) or (b_n) is a zero sequence.

☒ True

☐ False

◀ Previous

Not saved

false

Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 7 1 pts

Provided that it has at least two elements, the convergence set of the real power series $\sum_{n=0}^{\infty} a_n \cdot (x - x_0)^n$ is:

☒ an interval centered at x_0

☐ a closed interval centered at x_0

☒ not always an interval

☐ an open interval centered at x_0

Questions

- ✓ Question 1
- ✓ Question 2
- ✓ Question 3
- ✓ Question 4
- ✓ Question 5
- ✓ Question 6
- ✓ Question 7
- ⚙ Question 8

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
19 Minutes, 51 Seconds

Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 12 1 pts

If $(a_n + b_n)$ is a zero sequence, then either (a_n) or (b_n) is a zero sequence.

☐ True

☒ False

Questions

- ✓ Question 1
- ✓ Question 2
- ✓ Question 3
- ✓ Question 4
- ✓ Question 5
- ✓ Question 6
- ✓ Question 7
- ✓ Question 8

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
18 Minutes, 21 Seconds

Not saved Submit Quiz

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692996

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 5 1 pts

5. Give a numerical series which is convergent, but not absolutely convergent.

A) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ B) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$

C) $\sum_{n=1}^{\infty} \frac{-1}{2n}$ D) $\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n}$

☒ B)

☐ D)

☐ A)

☐ C)

Time Running: Hide
Attempt due: Jun 13 at 3:55pm
12 Minutes, 16 Seconds

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ENG 3:42 PM 6/13/2023

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692995

Question 3 1 pts

4. State the Cauchy convergence test for sequences.

A) A real number sequence is a Cauchy-sequence if and only if it is convergent.

B) A sequence $(a_n) : \mathbb{N} \rightarrow \mathbb{R}$ is convergent if and only if $\lim(\sqrt[n]{|a_n|}) < 1$

C) A real number sequence is a Cauchy-sequence if and only if it has limit.

☒ D) A real number sequence (a_n) is convergent if and only if

$$\exists N \in \mathbb{N} \forall \varepsilon > 0 \forall m, n \geq N : |a_n - a_m| < \varepsilon$$

☒ D)

☐ A)

☐ C)

☐ B)

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Search

ENG 3:08 PM 6/13/2023

a

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692993

Only one question is shown at a time.

Question 4 1 pts

Let (a_n) be a zero sequence, (b_n) be a convergent sequence, $\lim(b_n) = 2$. Then

☐ $\lim(a_n b_n) = 0$

☒ $\lim(a_n b_n) = 0$

☐ $\lim(a_n b_n) = -1$

☐ $\lim(a_n b_n) = 1$

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Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692997

Question 6 1 pts

Let $q \in \mathbb{K}$, $|q| < 1$. Then

A) $\sum_{n=0}^{\infty} q^n = \frac{q}{1-q}$

B) $\sum_{n=1}^{\infty} q^n = \frac{1}{1-q}$

C) $\sum_{n=0}^{\infty} q^n = \frac{1}{(1-q)^2}$

D) $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$

☐ A)

☐ B)

☒ D)

☐ C)

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
38 Minutes, 41 Seconds

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Analysis-1 exam (21 06 2022)

Basic level (15 questions)

Started: Jun 21 at 2:05pm

Quiz Instructions

In questions 1-10 mark the single correct answer

In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.

Ratio Test

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{\frac{1}{(n+1)^4}}{\frac{1}{n^4}} = \frac{n^4}{(n+1)^4} \approx 1$$

$$a_n = \frac{1}{n^4}$$

Ratio Test

Question 14

1 pts

The convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ cannot be decided with the Ratio Test.

☒ Igaz

turne true

☐ Hamis

false

◀ Previous

Next ▶

Not saved

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Questions

- ✓ Question 8
- ✓ Question 9
- ✓ Question 10
- ✓ Question 11
- ✓ Question 12
- ✓ Question 13

Quiz: Basic level of the exam (13)

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693139

Quiz Instructions

Only one ✓ You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Question 14 1 pts

If the terms of an infinite series converge to zero, then the series is convergent.

☐ False ^{1^n}

☒ True

◀ Previous Next ▶

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Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 15 Seconds

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2022/23/2

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Basic level of the exam (20 June 2023) (15 questions)

Started: Jun 20 at 3:06pm

Quiz Instructions

Question 12 1 pts

The convergence set of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \cdot (x-1)^n$ is the open interval $(0, 2)$.

☐ True

☒ False

◀ Previous Next ▶

No new data to save. Last checked at 3:40pm Submit Quiz

Questions

- ✓ Question 1
- ✓ Question 2
- ✓ Question 3
- ✓ Question 4
- ✓ Question 5
- ✓ Question 6
- ✓ Question 7
- ✓ Question 8

Time Running: Hide
Attempt due: Jun 20 at 3:50pm
9 Minutes, 54 Seconds

Activate Windows
Go to Settings to activate Windows.



Question 2

1 pts

Let $\emptyset \neq A \subseteq \mathbb{R}$, and $c = \sup A < +\infty$.

Which of the following statements is true?

- ☒ $\forall a \in A: a < c + 1$
- ☐ $\forall a \in A: a > c - 1$
- ☐ $\exists a \in A: a < c - 1$
- ☐ $\exists a \in A: a > c + 1$

sup

$$\forall a \in A: a \leq c$$

$$a < c+1$$

+1, inf

c = sup

c+1

c+p

Quiz: Basic level of the exam (20 x)

Started: Jun 20 at 3:06pm

Quiz Instructions

Question 11 1 pts

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. f is called continuous at the point a , if

- A) $\forall \varepsilon > 0 \forall \delta > 0 \forall x \in D_f, |x - a| < \delta: |f(x) - f(a)| < \varepsilon$
- ☒ B) $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D_f, |x - a| < \delta: |f(x) - f(a)| < \varepsilon$
- C) $\exists \varepsilon > 0 \forall \delta > 0 \forall x \in D_f, |x - a| < \delta: |f(x) - f(a)| < \varepsilon$
- D) $\exists \varepsilon > 0 \exists \delta > 0 \forall x \in D_f, |x - a| < \delta: |f(x) - f(a)| < \varepsilon$

Time Running: Hide
Attempt due: Jun 20 at 3:50pm
31 Minutes, 51 Seconds

Activate Windows
Go to Settings to activate Windows.



Question 12

1 pts

The convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ cannot be decided with the Ratio Test.

- ☐ True
- ☒ False

Quiz: Basic level of the exam (20 June 2023) (15 questions)

2022/23/2

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Basic level of the exam (20 June 2023) (15 questions)

Started: Jun 20 at 3:06pm

Quiz Instructions

Question 13 1 pts

Let $f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. Then f is continuous at the point a if and only if for any sequence $x_n \in D_f$ ($n \in \mathbb{N}$) holds $\lim_{n \rightarrow \infty} f(x_n) = f(a)$.

☒ False
☐ True

Previous Next

Quiz saved at 3:41pm Submit Quiz

Questions

- ✓ Question 8
- ✓ Question 9
- ✓ Question 10
- ✓ Question 11
- ✗ Question 12
- ✓ Question 13
- ✓ Question 14
- ✓ Question 15

Time Running: [Hide](#)
Attempt due: Jun 20 at 3:50pm
8 Minutes, 34 Seconds

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30°C
Tâmbayre napos

Search

ENG 15:41
20.06.2023

true

Quiz: Basic level of the exam (13 June 2023) (8 questions)

2022/23/2

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Basic level of the exam (13 June 2023) (8 questions)

Only one question is shown at a time.

Quiz Instructions

Question 2 1 pts

4. Which of the following statements is true?

A) If a real number sequence is bounded above, then it is convergent.
B) If a real number sequence is bounded, then it is convergent.
C) If a real number sequence is convergent, then it is bounded above.
D) If a real number sequence is convergent, then it is not bounded below.

☐ C
☐ B
☒ A
☐ D

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No new data to save. Last checked at 3:07pm Submit Quiz

Questions

- ① Question 3
- ② Question 4
- ③ Question 5
- ④ Question 6
- ⑤ Question 7
- ⑥ Question 8

Time Running: [Hide](#)
Attempt due: Jun 13 at 3:50pm
42 Minutes, 44 Seconds

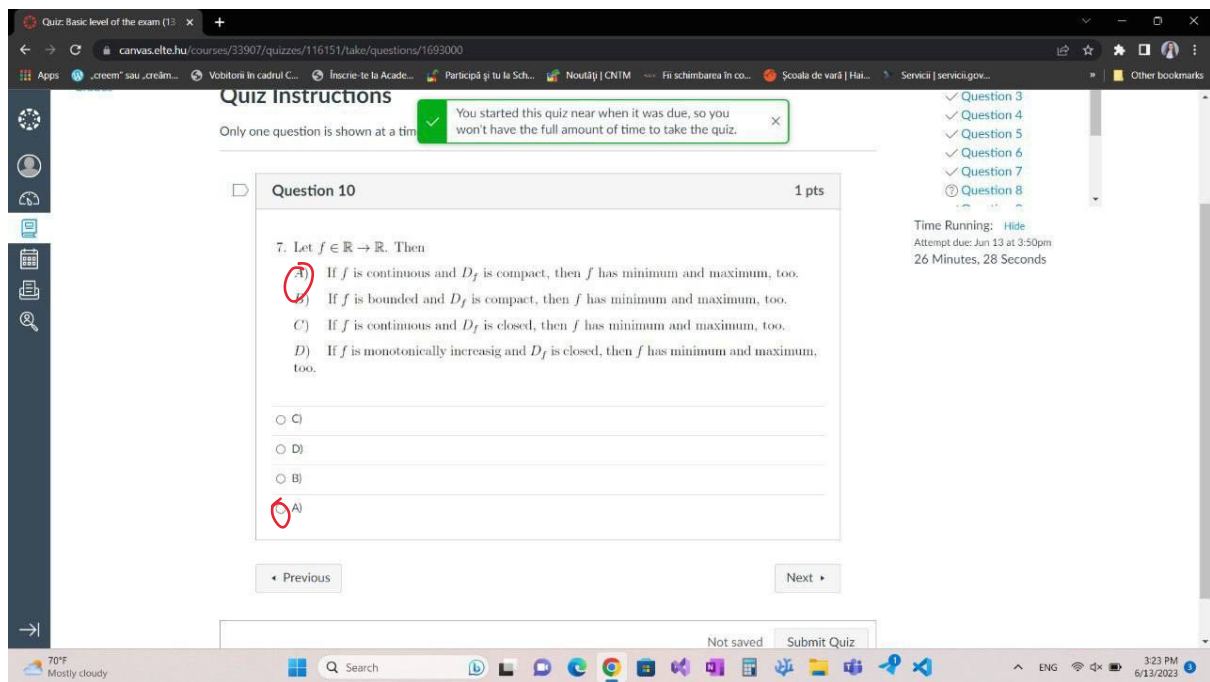
68°F
Mostly cloudy

Search

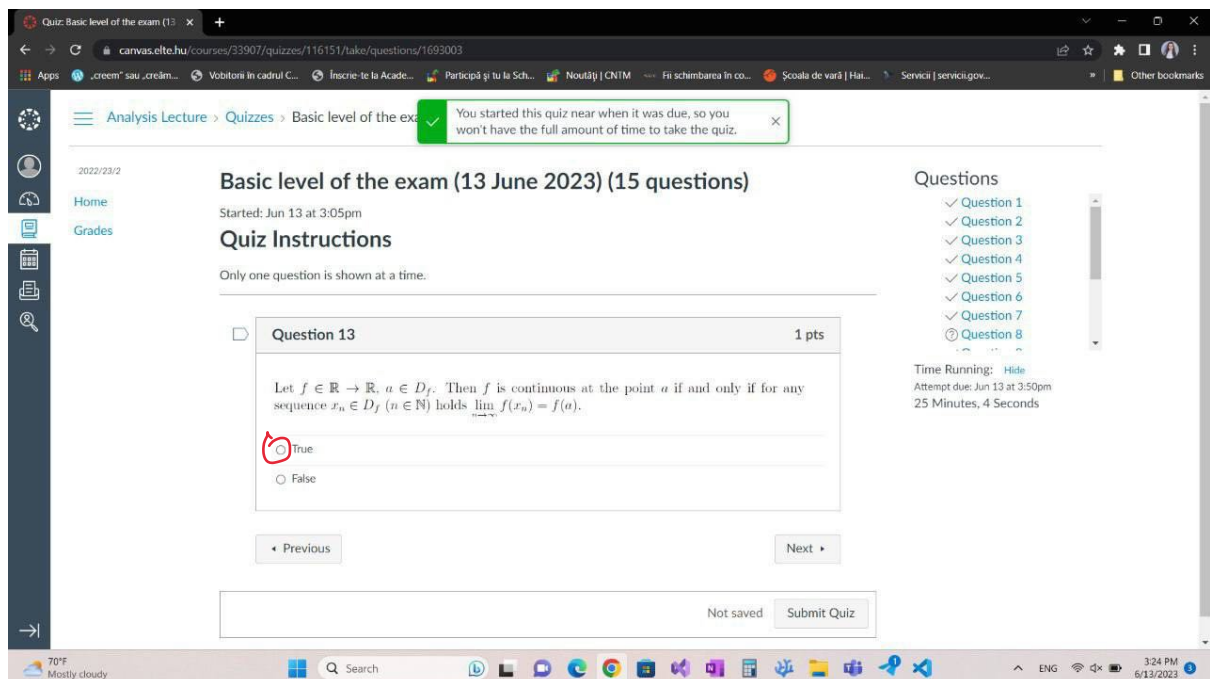
ENG 3:07 PM
6/13/2023

C





a



false

Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 15 1 pts

15. If a function of type $\mathbb{R} \rightarrow \mathbb{R}$ is continuous, then it has minimum and it has maximum.

☐ True

☒ False

FALSE

◀ Previous

Questions

- ✓ Question 1
- ✓ Question 2
- ✓ Question 3
- ✓ Question 4
- ✓ Question 5
- ✓ Question 6
- ✓ Question 7
- ⓘ Question 8

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
20 Minutes, 59 Seconds

Quiz saved at 3:28pm Submit Quiz

false

Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

Quiz Instructions

Only one question is shown at a time.

Question 14 1 pts

9. If a nonempty real number set is bounded below, then it has minimum.

☐ True

☒ False

Previous Next

Questions

- ✓ Question 1
- ✓ Question 2
- ✓ Question 3
- ✓ Question 4
- ✓ Question 5
- ✓ Question 6
- ✓ Question 7
- ⓘ Question 8

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
22 Minutes, 24 Seconds

No new data to save. Last checked at 3:27pm Submit Quiz



Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693145

Only one question is shown at a time.

Question 2 1 pts

Let $a_n, b_n, c_n \in \mathbb{R} \ n \in \mathbb{N}$.

A) If $\lim(b_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(a_n) = A$

☒ B) If $\forall n \in \mathbb{N} : a_n \leq b_n \leq c_n$ and $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

C) If $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

D) If $\lim(a_n) = \lim(b_n) = A \in \mathbb{R}$, then $\lim(c_n) = A$

☐ C)

☒ B)

☐ A)

☐ D)

Previous Next

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
43 Minutes, 23 Seconds

Activate Windows
Go to Settings to activate Windows.

29°C Zápork



Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693147

Quiz

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Only one question is shown at a time.

Question 7 1 pts

4. Which of the following statements is true?

A) If $\sum a_n$ is convergent, then $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$

B) If a series is convergent then it is absolutely convergent.

C) If $\lim(a_n) = 0$, then $\sum a_n$ is convergent.

☒ D) If $\sum a_n$ is convergent, then $\lim(a_n) = 0$.

☐ C)

☐ B)

☒ D)

☐ A)

Previous Next

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
31 Minutes, 35 Seconds

Activate Windows
Go to Settings to activate Windows.

29°C Zápork



Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Quiz Instructions

Only one question is shown at a time.

Question 12 1 pts

9. If a nonempty real number set is bounded below, then it has minimum.

☐ True

☒ False

Previous Next

Not saved Submit Quiz

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 37 Seconds

Activate Windows
Go to Settings to activate Windows.



Quiz: Basic level of the exam (13 June 2023) (15 questions)

Started: Jun 13 at 3:05pm

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Basic level of the exam (13 June 2023) (15 questions)

Quiz Instructions

Only one question is shown at a time.

Question 13 1 pts

11. If the limit of a number sequence is $-\infty$, then it is not bounded below.

☒ True

☐ False

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 28 Seconds

Activate Windows
Go to Settings to activate Windows.

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693139

Quiz Instructions

Only one question is shown at a time.

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Question 14 1 pts

If the terms of an infinite series converge to zero, then the series is convergent.

☒ False

☐ True

Previous Next

Not saved Submit Quiz

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 15 Seconds

Activate Windows
Go to Settings to activate Windows.

29°C Zápork

Search

3:19 PM 6/13/2023



Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693145

Only one question is shown at a time.

Question 2 1 pts

Let $a_n, b_n, c_n \in \mathbb{R} \ n \in \mathbb{N}$.

A) If $\lim(b_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(a_n) = A$

☒ B) If $\forall n \in \mathbb{N} : a_n \leq b_n \leq c_n$ and $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

C) If $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

D) If $\lim(a_n) = \lim(b_n) = A \in \mathbb{R}$, then $\lim(c_n) = A$

☐ C)

☒ B)

☐ A)

☐ D)

Previous Next

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
43 Minutes, 23 Seconds

Activate Windows
Go to Settings to activate Windows.

29°C Zápork

Search

3:06 PM 6/13/2023

Round 2



Question 1

1 pts

5. Choose the statement which is true :

- (a) If $\lim(a_n \cdot b_n) = 0$, then either (a_n) or (b_n) is a zero sequence.
- (b) If (a_n) is a zero sequence and $(b_n) : \mathbb{N} \rightarrow (-\infty; 0)$ is a convergent sequence, then $\lim \left(\frac{a_n}{b_n} \right) = 0$.
- (c) There is no such positive term zero sequence (a_n) and a divergent sequence (b_n) for which holds that $(a_n \cdot b_n)$ is a zero sequence.
- (d) Let (a_n) be a positive term sequence for which holds that $\forall P > 0$ the set $\{n \in \mathbb{N} : a_n \notin (P; +\infty)\}$ is a finite set. Then $\left(\frac{1}{a_n} \right)$ is a zero sequence.

☐ A)

☐ B)

☐ C)

☒ D)

The number sequence (a_n) is alternating (that is $a_{n+1} = a_n$ for all $n \in \mathbb{N}$), and the sequence $(|a_n|)$ is strictly monotonically decreasing. Then

☒ (a) (a_n) is bounded

☐ (a_n) is a zero sequence

☐ (a_n) is convergent

☐ (a_n) is divergent

lim has to be = 0
to be convergent

answer is a

convergent



Question 3

1 pts

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, and suppose that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and that $\lim_{x \rightarrow -\infty} f(x) = -\infty$. Which of the following statements is true?

- A) The function f is monotonically increasing.
- B) The equation $f(x) = 0$ has at least one solution in the set of real numbers.
- C) There exists an example for f such that the range of f is not the whole set of real numbers (that is: $R_f \neq \mathbb{R}$).
- D) The range of f is always $\mathbb{R}$.

☒ A)

☐ B)

☐ C)

☐ D)

C



Question 4

1 pts

4. Choose the statement which is true :

a hand
written

- (a) Let $\emptyset \neq A \subset \mathbb{R}$ and suppose that $\forall \varepsilon > 0 \exists a \in A : a < 2 - \varepsilon$. Then $\sup A = 2$.
- (b) There exists a set $A \subset \mathbb{R}$ which has at least two elements, and for which holds $\inf A = \sup A$.
- (c) Let $\emptyset \neq A \subset \mathbb{R}$ be a bounded set. Then $\sup (A \cup (0, 1)) = \max\{\sup A; 1\}$.
- (d) There is no such nonempty and bounded set $\emptyset \neq A \subset \mathbb{R}$, for which holds $\forall \varepsilon > 0 \forall a \in A : a > \sup A - \varepsilon$.

☒ A)

☐ B)

☒ C)

☐ D)

C



Question 5

1 pts

10. Choose the statement which is true :

- (a) Let $f \in \mathbb{R} \rightarrow \mathbb{R}, a \in D_f$. If $f \in C\{a\}$, then $\frac{1}{f} \in C\{a\}$.
- (b) There exists such a continuous function $f : (0, 2] \rightarrow \mathbb{R}$, for which hold $f(1) \cdot f(2) < 0$ and $f(x) < 0$ for all $x \in (0; 2]$.
- (c) Suppose that $f \in \mathbb{R} \rightarrow \mathbb{R}$ and $f \in C\{a\}$. Then $\exists \lim_a (f \circ f) = f(f(a))$.
- (d) $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in \mathbb{R} : |x| < \delta : |\exp(x) - 1| < \varepsilon$.

Correct answer

☒ A)

☐ B)

☐ C)

☐ D)

d



Question 6

1 pts

2. Choose the statement which is **false** :

- ☒ (a) If the sequence (a_n) is divergent then (a_n) is not bounded. **FALSE**
- (b) If the limit of a real sequence (a_n) is $+\infty$, then (a_n) is divergent.
- (c) There is no such real sequence (a_n) , which is unbounded below and whose limit is $+\infty$.
- (d) If the real sequence (a_n) is not bounded, then (a_n) is divergent.

☐ A)

☐ B)

☐ C)

☐ D)

a



Question 6

1 pts

2. Choose the statement which is **false** :

- (a) If the sequence (a_n) is divergent then (a_n) is not bounded.
- (b) If the limit of a real sequence (a_n) is $+\infty$, then (a_n) is divergent.
- (c) There is no such real sequence (a_n) , which is unbounded below and whose limit is $+\infty$.
- (d) If the real sequence (a_n) is not bounded, then (a_n) is divergent.

☒ A)

☐ B)

☐ C)

☐ D)



Question 1

1 pts

10. Choose the statement which is true :

- (a) Let $f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. If $f \in C\{a\}$, then $\frac{1}{f} \in C\{a\}$.
- (b) There exists such a continuous function $f : (0, 2] \rightarrow \mathbb{R}$, for which hold $f(1) \cdot f(2) < 0$ and $f(x) < 0$ for all $x \in (0, 2]$.
- (c) Suppose that $f \in \mathbb{R} \rightarrow \mathbb{R}$ and $f \in C\{a\}$. Then $\exists \lim_x (f \circ f) = f(f(a))$.
- (d) $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in \mathbb{R} : |x| < \delta : |\exp(x) - 1| < \varepsilon$.

☐ A)

☐ B)

d

16:18



You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

June 2023) (7 quiz question



Account



Dashboard



Courses



Calendar



Inbox



StudyCoach

Jun 20 at 4:16pm

Instructions

Question 7

1 pts

The number sequence (a_n) is alternating (that is $a_{n+1} = -a_n$ for all $n \in \mathbb{N}$), and the sequence $(|a_n|)$ is strictly monotonically decreasing. Then

- ☒ (a_n) is bounded
- ☐ (a_n) is a zero sequence
- ☐ (a_n) is convergent
- ☐ (a_n) is divergent

◀ Previous

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Advanced level Part 1 of the  

7/quizzes/116316/take/questions/1698814

be  Maps  Translate  PowerPoint Macros  Roadmap  Difference between... FouadKh219/Ab

Started: Jun 20 at 4:11pm

Quiz Instructions

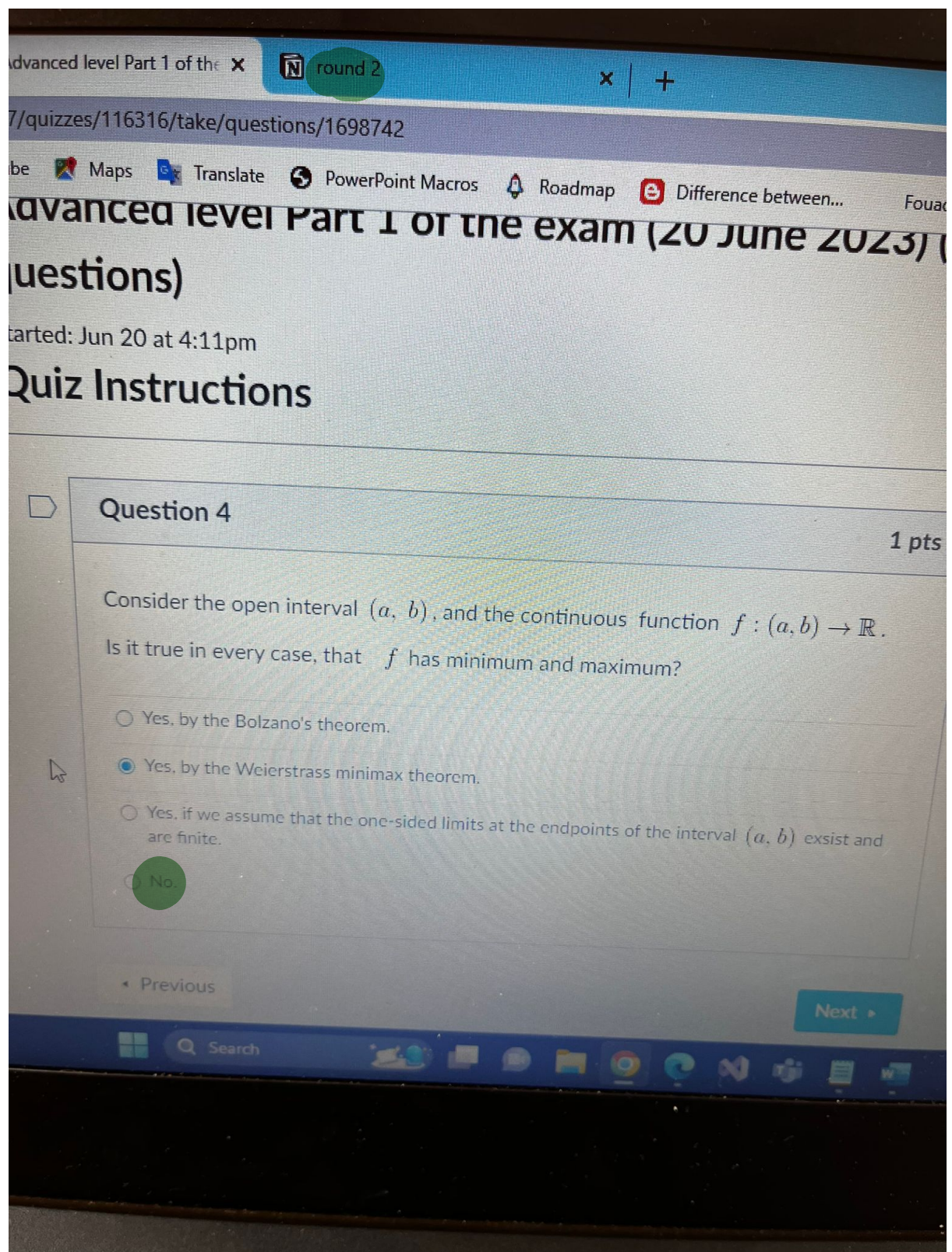
Question 3

1 pts

10. Choose the statement which is true :

- (a) Let $f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. If $f \in C^1\{a\}$, then $\frac{1}{f} \in C^1\{a\}$.
- (b) There exists such a continuous function $f : (0, 2] \rightarrow \mathbb{R}$, for which hold $f(1) \cdot f(2) < 0$ and $f(x) < 0$ for all $x \in (0, 2]$.
- (c) Suppose that $f \in \mathbb{R} \rightarrow \mathbb{R}$ and $f \in C^1\{a\}$. Then $\exists \lim_{a} (f \circ f) = f(f(a))$.
- (d) $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in \mathbb{R} : |x| < \delta : |\exp(x) - 1| < \varepsilon$.

☐ A)
☐ B)
☐ C)
☒ D)



bolzano's



Question 1

1 pts

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, and suppose that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and that $\lim_{x \rightarrow -\infty} f(x) = -\infty$. Which of the following statements is true?

- A) The function f is monotonically increasing.
- B) The equation $f(x) = 0$ has at least one solution in the set of real numbers.
- C) There exists an example for f such that the range of f is not the whole set of real numbers (that is: $R_f \neq \mathbb{R}$).
- D) The range of f is always $\mathbb{R}$.

☒ A)

☐ B)

☐ C)

☐ D)

Next >



Search



1 pts

(a) If the sequence (a_n) is divergent then (a_n) is not bounded.

(b) If the limit of a real sequence (a_n) is $-\infty$, then (a_n) is divergent.

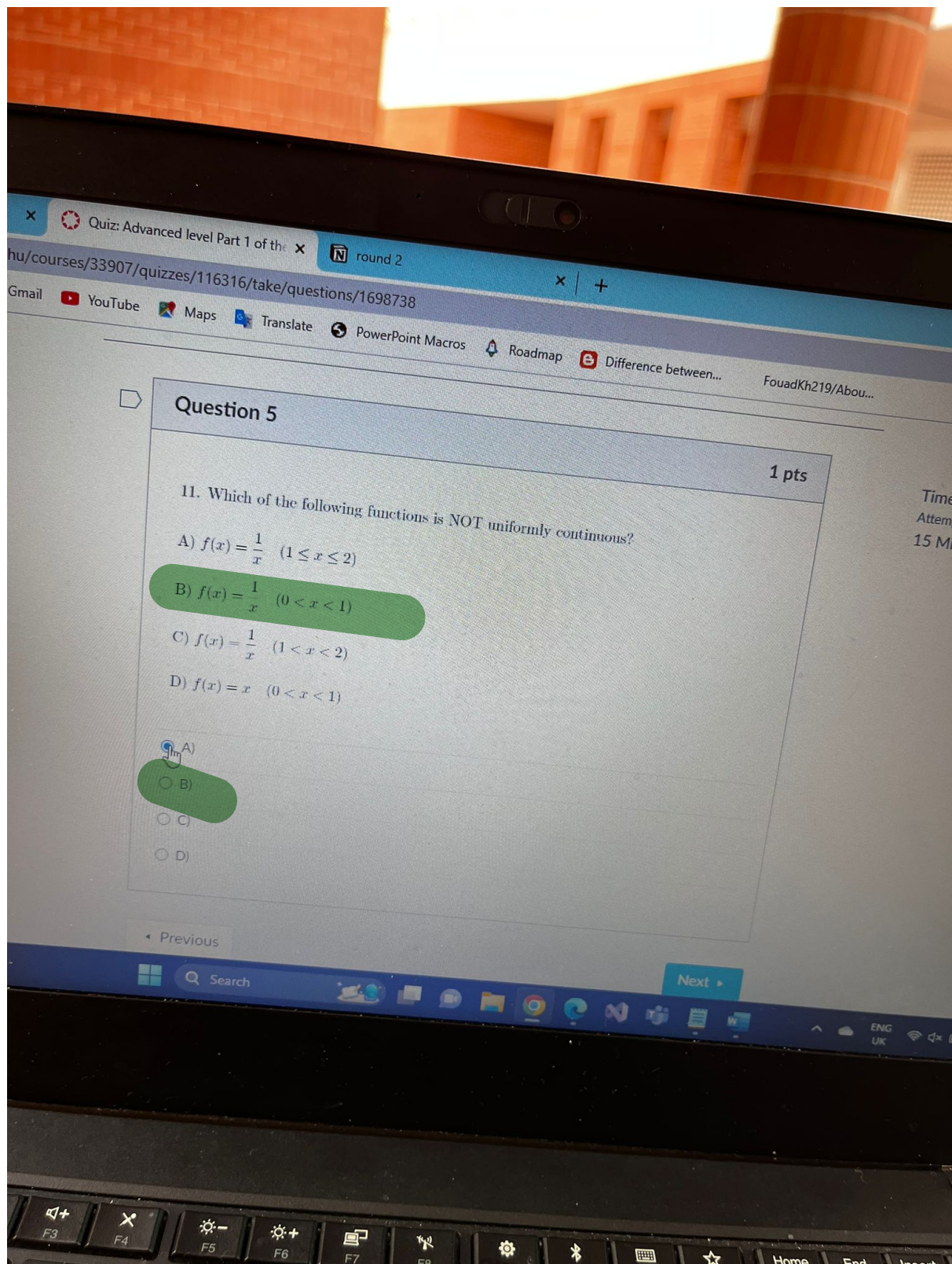
(c) There is no such real sequence (a_n) , which is unbounded below and whose limit is $+\infty$.

(d) If the real sequence (a_n) is not bounded, then (a_n) is divergent.

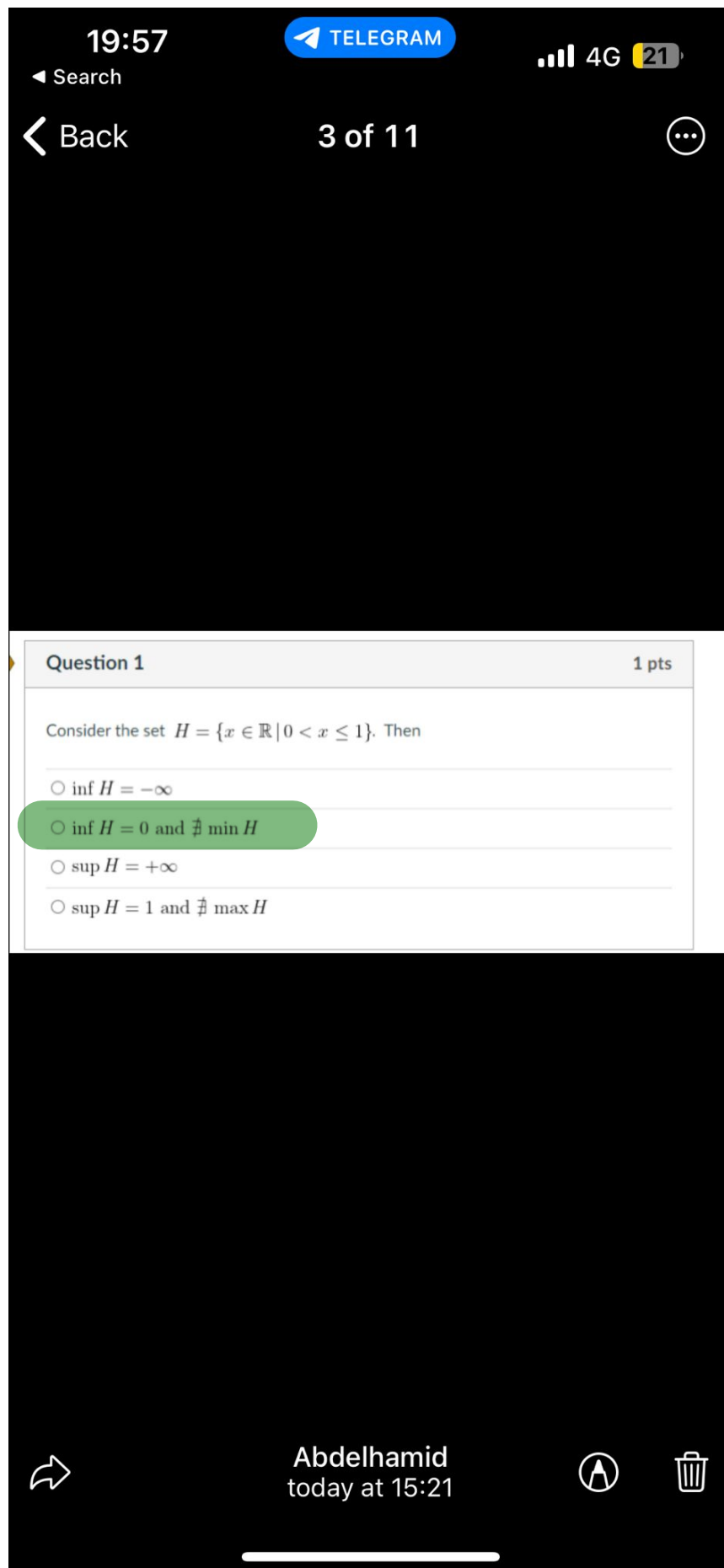
☐ B)

☐ D)





b



b

Advanced level Part 1 of the x round 2

07/quizzes/116316/take/questions/1698813

Tube Maps Translate PowerPoint Macros Roadmap Difference between... FouadKh219/Abou...

Quiz instructions

Question 6 1 pts

8. Which of the following power series has the convergence set $(-1, 1]$?

(a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$

(c) $\sum_{n=1}^{\infty} x^n$

(d) $\sum_{n=1}^{\infty} (-1)^n x^n$

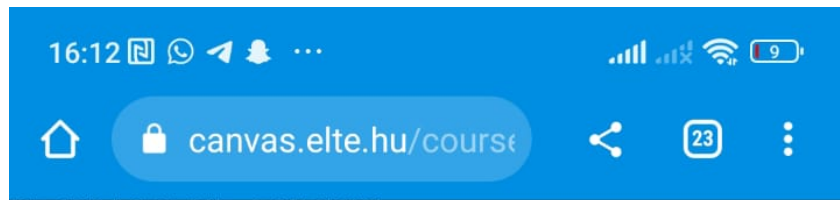
☐ A)

☒ B)

☐ C)

☐ D)

Search



iz Instructions

Question 2

1

7. Choose the statement which is true :

- (a) If $\sum(a_n)$ is a convergent positive term series, then $\lim \left(\frac{a_{n+1}}{a_n} \right) < 1$.
- (b) If $\lim(|a_n|) = 0$, then $\sum(a_n)$ is absolutely convergent. *False*
- (c) If $\sum_{n=1}^{\infty} (a_n)$ is convergent, then $\sum_{n=1}^{\infty} \left(\frac{n \cdot a_n + 1}{n} \right)$ is also convergent.
- (d) If the positive term series $\sum(a_n)$ is convergent, then the series $\sum((-1)^n \cdot a_n)$ is also convergent.

☐ A)

☒ B)

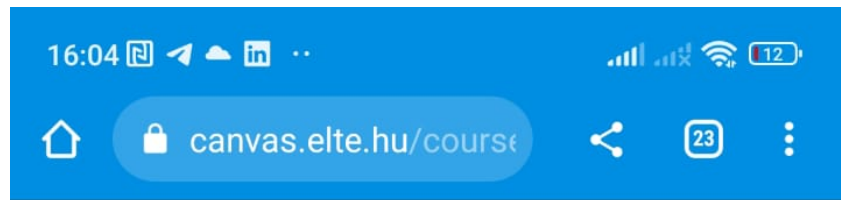
☐ C)

☐ D)

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Advanced level Part 1 of the exam (0 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions

Question 7

1 pts

8. Which of the following power series has the convergence set $(-1, 1]$?

(a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$

(c) $\sum_{n=1}^{\infty} x^n$

(d) $\sum_{n=1}^{\infty} (-1)^n x^n$

☐ A)

☐ B)

☐ C)

☐ D)

◀ Previous

Not saved

Submit Quiz

b

quiz near when it was due, so you
full amount of time to take the quiz.

× Part 1 of t...

Advanced level Part 1 of the exam (20 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions



Question 6

1 pts

4. Choose the statement which is true :

- (a) Let $\emptyset \neq A \subset \mathbb{R}$ and suppose that $\forall \varepsilon > 0 \exists a \in A : a < 2 - \varepsilon$. Then $\sup A = 2$.
- (b) There exists a set $A \subset \mathbb{R}$ which has at least two elements, and for which holds $\inf A = \sup A$.
- (c) Let $\emptyset \neq A \subset \mathbb{R}$ be a bounded set. Then $\sup(A \cup (0, 1)) = \max\{\sup A; 1\}$.
- (d) There is no such nonempty and bounded set $\emptyset \neq A \subset \mathbb{R}$, for which holds $\forall \varepsilon > 0 \forall a \in A : a > \sup A - \varepsilon$.

☐ A)

☐ B)

☐ C)

☐ D)

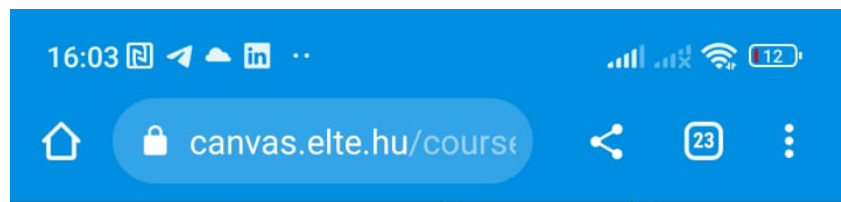
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Submit Quiz

C



Advanced level Part 1 of the exam (20 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions



Question 5

1 pts

Let $\emptyset \neq A \subseteq \mathbb{R}$.

Which of the following statements is true?

- ☒ If A has maximum and it has finite infimum, then A is bounded set.
- ☐ If A has finite infimum, then it has minimum.
- ☐ If A is bounded, then A has maximum.
- ☐ If A has neither minimum nor maximum, then A is not bounded.

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Not saved

Submit Quiz

a

Advanced level Part 1 of the exam (20 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions



Question 3

1 pts

Consider the open interval (a, b) , and the continuous function $f : (a, b) \rightarrow \mathbb{R}$.

Is it true in every case, that f has minimum and maximum?

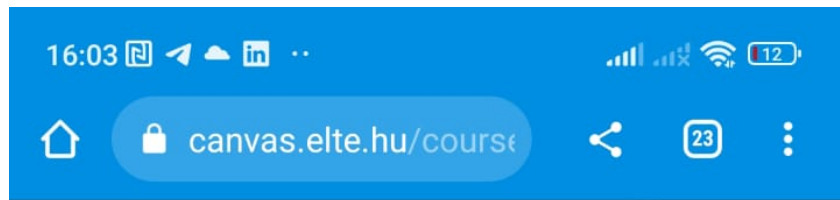
- ☒ Yes, by the Bolzano's theorem.
- ☐ Yes, by the Weierstrass minimax theorem.
- ☐ Yes, if we assume that the one-sided limits at the endpoints of the interval (a, b) exist and are finite.
- ☐ No.

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No new data to save. Last checked at 4:14pm

Submit Quiz



Advanced level Part 1 of the exam (0 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions

Question 2

1 pts

7. Choose the statement which is true :

(a) If $\sum(a_n)$ is a convergent positive term series, then $\lim \left(\frac{a_{n+1}}{a_n} \right) < 1$.

(b) If $\lim(|a_n|) = 0$, then $\sum(a_n)$ is absolutely convergent.

(c) If $\sum(a_n)$ is convergent, then $\sum \left(\frac{n \cdot a_n + 1}{n} \right)$ is also convergent.

(d) If the positive term series $\sum(a_n)$ is convergent, then the series $\sum((-1)^n \cdot a_n)$ is also convergent.

☐ A)

☐ B)

☐ C)

☒ D)

◀ Previous

Next ▶

Not saved

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unsure

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

× Part 1 of t...

Advanced level Part 1 of the exam (20 June 2023) (7 quiz questions)

Started: Jun 20 at 4:03pm

Quiz Instructions



Question 1

1 pts

Let (z_n) be a zero sequence, (d_n) be a divergent sequence. Then

- ☐ $(z_n \cdot d_n)$ is a zero sequence
- ☐ $(z_n \cdot d_n)$ is a bounded sequence
- ☐ the sequence $(z_n + d_n)$ is not bounded
- ☒ the sequence $(z_n + d_n)$ is divergent

} FALSE

Next ▶

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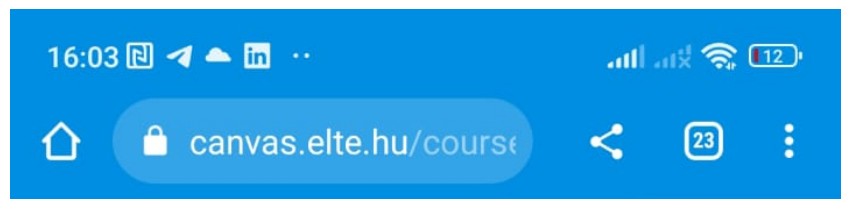
Submit Quiz

Questions

- ? Question 1
- ? Question 2
- ? Question 3
- ? Question 4
- ? Question 5
- ? Question 6
- ? Question 7

Time Running: [Hide](#)

d



quiz near when it was due, so you
amount of time to take the quiz.

× Part 1 of t...

Advanced level Part 1 of the exam (0 June 2023) (7 quiz questions)

Created: Jun 20 at 4:03pm

Quiz Instructions

Question 4

1 pts

2. Choose the statement which is **false** :

- (a) If the sequence (a_n) is divergent then (a_n) is not bounded.
- (b) If the limit of a real sequence (a_n) is $+\infty$, then (a_n) is divergent.
- (c) There is no such real sequence (a_n) , which is unbounded below and whose limit is $+\infty$.
- (d) If the real sequence (a_n) is not bounded, then (a_n) is divergent.

☐ A)

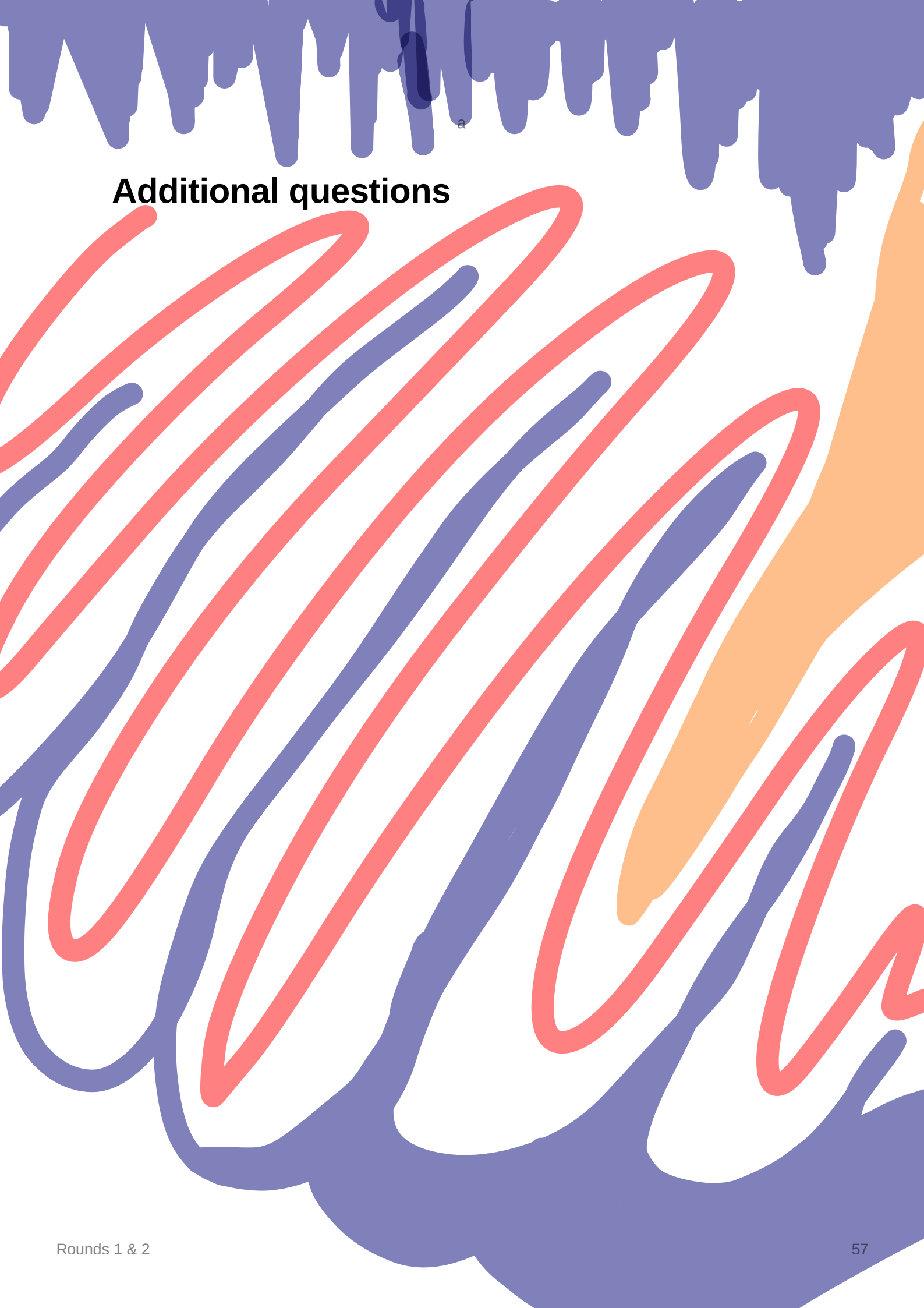
☐ B)

☐ C)

☐ D)

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Next ▶



Additional questions



In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.



Question 5

1 pts

Let $q \in \mathbb{K}$, $|q| < 1$. Then

A) $\sum_{n=0}^{\infty} q^n = \frac{q}{1-q}$

B) $\sum_{n=1}^{\infty} q^n = \frac{1}{1-q}$

C) $\sum_{n=0}^{\infty} q^n = \frac{1}{(1-q)^2}$

D) $\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$

☐ A)

☐ B)

☐ C)

☒ D)

$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$

14:20

TELEGRAM

Quiz: Analysis-1 exam...
canvas.elte.hu

In questions 1-10 mark the single correct answer

In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.



Question 8

1 pts

Choose the correct answer for

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{6x} = ?$$

☐ 1☐ 2☒ $\frac{1}{2}$ ☐ 0

$$\frac{3}{6} = \frac{1}{2}$$

◀ Previous

Next ▶

Not saved

Submit Quiz



Question 9

6. Define the sin function.

A) $\sin(x) := \sum_{n=1}^{+\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$

B) $\sin(x) := \sum_{n=0}^{+\infty} \frac{(-x)^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$

C) $\sin(x) := \sum_{n=0}^{+\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$

D) $\sin(x) := \sum_{n=0}^{+\infty} \frac{x^{2n+1}}{(2n+1)!} \quad (x \in \mathbb{R}).$

☐ A)

☐ B)

☒ C)

☐ D)

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

◀ Previous

14:20

TELEGRAM

Quiz: Analysis-1 exam...
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✕

🔒

🔗

🔖

⋮

false

Only one question is shown at a time.

Question 6

1 pts

6. Which of the following series is conditionally convergent?

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

$$\sum_{n=1}^{\infty} \frac{-1}{2n}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+1}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n}$$

☒ A)

☐ B)

☐ C)

☐ D)

◀ Previous

Next ▶

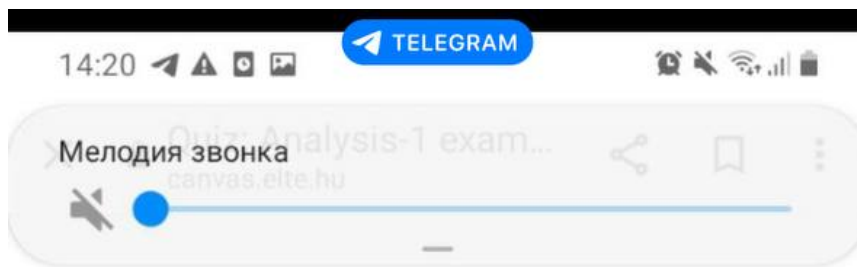
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Questions

Rounds 1 & 2

61



the question is shown at a time.

Question 7**1 pts**

Provided that it has at least two elements, the convergence set of the real power series $\sum_{n=0}^{\infty} a_n \cdot (x - x_0)^n$ is:

- ☐ an open interval centered at x_0
- ☒ an interval centered at x_0
- ☐ a closed interval centered at x_0
- ☐ not always an interval

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not always an interval

one question is shown at a time.

Question 11

9. If a real number set has minimum, then it is bounded above.

☐ True

☒ False

not necessarily is
above,
but it's bounded
below

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Analysis-1 exam (21 06 2022)

Basic level (15 questions)

Started: Jun 21 at 2:05pm

Quiz Instructions

In questions 1-10 mark the single correct answer

In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.



Question 14

1 pts

The convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ cannot be decided with the Ratio Test.

☒ Igaz

☐ Hamis

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Questions

- ✓ Question 8
- ✓ Question 9
- ✓ Question 10
- ✓ Question 11
- ✓ Question 12
- ✓ Question 13

one question is shown at a time.

Question 6

6. Which of the following series is conditionally convergent?

A) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

B) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$

C) $\sum_{n=1}^{\infty} \frac{-1}{2n}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n}$

☒ A)

☐ B)

☐ C)

☐ D)

◀ Previous

14:20

TELEGRAM

14:20

Quiz: Analysis-1 exam...

canvas.elte.hu

my one question is shown at a time.

Question 13

1 pts

If the terms of an infinite series converge to zero, then the series is convergent.

false

☒ Igaz

☐ Hamis

◀ Previous

Next ▶

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Questions

✓ Question 1

✓ Question 2

✓ Question 3

✓ Question 4

tions 11-15 mark that the statement is

the question is shown at a time.

Question 3

1

4. Which of the following statements is true?

- A) If a real number sequence is bounded above, then it is convergent.
- B) If a real number sequence is bounded, then it is convergent.
- C) If a real number sequence is convergent, then it is bounded above.
- D) If a real number sequence is convergent, then it is not bounded below.

☐ A)

☐ B)

☒ C)

☐ D)

◀ Previous

Next ▶



In questions 1-10 mark the single correct answer

In questions 11-15 mark that the statement is true or false

Only one question is shown at a time.



Question 4

4. State the Cauchy convergence test for sequences.

- A) A real number sequence is a Cauchy-sequence if and only if $\lim_{n \rightarrow \infty} a_n$ exists.
- B) A sequence $(a_n) : \mathbb{N} \rightarrow \mathbb{R}$ is convergent if and only if $\lim_{n \rightarrow \infty} a_n$ exists.
- C) A real number sequence is a Cauchy-sequence if and only if $\lim_{n \rightarrow \infty} a_n$ exists.
- D) A real number sequence (a_n) is convergent if and only if $\exists N \in \mathbb{N} \forall \varepsilon > 0 \forall m, n \geq N : |a_m - a_n| < \varepsilon$.

$$\exists N \in \mathbb{N} \forall \varepsilon > 0 \forall m, n \geq N : |a_m - a_n| < \varepsilon$$

☐ A)

☐ B)

☐ C)

☒ D)

a

> **Question 1**

1 pts

Consider the set

$H = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$. Then

☒ $\inf H = 0$ and $\nexists \min H$

☐ $\inf H = -\infty$

☐ $\sup H = +\infty$

☐ $\sup H = 1$ and $\nexists \max H$

d

Question 8

1 pts

Choose the correct answer for

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{x} = ?$$

☐ $\frac{1}{7}$

☐ 1

☐ 0

☒ 7

Question 11

1 pts

11. If the limit of a number sequence is $-\infty$, then it is not bounded below.

☒ True

☐ False

Question 13

1 pts

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) \cdot f(b) < 0$, then $\exists c \in (a, b) : f(c) = 0$.

true

☒ False

☐ True

Question 14

1 pts

If $(a_n + b_n)$ is a zero sequence, then either (a_n) or (b_n) is a zero sequence.

☐ True

☒ False

Question 15

1 pts

Let $f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. Then f is continuous at the point a if and only if for any sequence $x_n \in D_f$ ($n \in \mathbb{N}$) holds $\lim_{n \rightarrow \infty} f(x_n) = f(a)$.

☒ True

☐ False



Only one question is shown at a time.



Question 10

1 pts

Let $\emptyset \neq H \subseteq \mathbb{R}$. The set H is open set if and only if

- A) Any point in it is interior point
- B) It has at least one interior point
- C) It has infinitely many interior points
- D) It has no boundary point

☒ A)

☐ B)

☐ C)

☐ D)

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Questions



Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693145

Only one question is shown at a time.

Question 2 1 pts

Let $a_n, b_n, c_n \in \mathbb{R}$ $n \in \mathbb{N}$.

A) If $\lim(b_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(a_n) = A$

B) If $\forall n \in \mathbb{N} : a_n \leq b_n \leq c_n$ and $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

C) If $\lim(a_n) = \lim(c_n) = A \in \mathbb{R}$, then $\lim(b_n) = A$

D) If $\lim(a_n) = \lim(b_n) = A \in \mathbb{R}$, then $\lim(c_n) = A$

☐ C)

☒ B)

☐ A)

☐ D)

Previous Next

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
43 Minutes, 23 Seconds

Activate Windows
Go to Settings to activate Windows.

29°C Zápork

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1692993

Only one question is shown at a time.

Question 4 1 pts

Let (a_n) be a zero sequence, (b_n) be a convergent sequence, $\lim(b_n) = 2$.
Then

☐

☒ $\lim(a_n b_n) = 0$

☐ $\lim(a_n b_n) = -1$

☐ $\lim(a_n b_n) = 1$

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Activate Windows
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29°C Zápork

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693147

Quiz

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Only one question is shown at a time.

Question 7 1 pts

4. Which of the following statements is true?

A) If $\sum a_n$ is convergent, then $\lim_{n \rightarrow +\infty} \sqrt[n]{|a_n|} < 1$

B) If a series is convergent then it is absolutely convergent.

C) If $\lim(a_n) = 0$, then $\sum a_n$ is convergent.

D) If $\sum a_n$ is convergent, then $\lim(a_n) = 0$.

☐ C)

☐ B)

☒ D)

☐ A)

Previous Next

Time Running: Hide
Attempt due: Jun 13 at 3:50pm
31 Minutes, 35 Seconds

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29°C Zápork

Quiz: Basic level of the exam (13) x

canvas.elte.hu/courses/33907/quizzes/116151/take/questions/1693140

Grades

Quiz instructions

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Only one question is shown at a time.

Question 11 1 pts

15. If a function of type $\mathbb{R} \rightarrow \mathbb{R}$ is continuous, then it has minimum and it has maximum.

☐ True

☒ False

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Time Running: Hide
Attempt due: Jun 13 at 3:50pm
30 Minutes, 45 Seconds

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29°C Zápork

Quiz: Basic level of the exam (13) x

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Quiz Instructions

Only one correct answer

You started this quiz near when it was due, so you won't have the full amount of time to take the quiz.

Question 14 1 pts

If the terms of an infinite series converge to zero, then the series is convergent.

☐ False

☒ True

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Time Running: [Hide](#)
Attempt due: Jun 13 at 3:50pm
30 Minutes, 15 Seconds

Question 5
Question 6
Question 7
Question 8

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Go to Settings to activate Windows.

29°C Záhporok
Search
3:19 PM 6/13/2023

Question 7

1 pts

5. Give a numerical series which is convergent, but not absolutely convergent.

A) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

B) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$

C) $\sum_{n=1}^{\infty} \frac{-1}{2n}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n}$

☐ D)

☒ A)

☐ B)

☐ C)

Question 9

1 pts

Let $f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in D_f$. f is called continuous at the point a , if

A) $\forall \varepsilon > 0 \forall \delta > 0 \forall x \in D_f, |x - a| < \delta : |f(x) - f(a)| < \varepsilon$

B) $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D_f, |x - a| < \delta : |f(x) - f(a)| < \varepsilon$

C) $\exists \varepsilon > 0 \forall \delta > 0 \forall x \in D_f, |x - a| < \delta : |f(x) - f(a)| < \varepsilon$

D) $\exists \varepsilon > 0 \exists \delta > 0 \forall x \in D_f, |x - a| < \delta : |f(x) - f(a)| < \varepsilon$

☐ A)

☐ C)

☐ D)

☒ B)

15:13



c level of the exam (13 June



3) (15 questions)



Jun 13 at 3:05pm



Instructions



e question is shown at a time.



Question 10

1 pts

7. Let $f \in \mathbb{R} \rightarrow \mathbb{R}$. Then

- A) If f is continuous and D_f is compact, then f has minimum and maximum, too.
- B) If f is bounded and D_f is compact, then f has minimum and maximum, too.
- C) If f is continuous and D_f is closed, then f has minimum and maximum, too.
- D) If f is monotonically increasing and D_f is closed, then f has minimum and maximum, too.

☐ B)☒ A)☐ D)☐ C)[◀ Previous](#)[Next ▶](#)

Quiz saved at 3:13pm

[Submit Q](#)canvas.elte.hu

Question 7

1 pts

5. Give a numerical series which is convergent, but not absolutely convergent.

A) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

B) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$

C) $\sum_{n=1}^{\infty} \frac{-1}{2n}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n}$

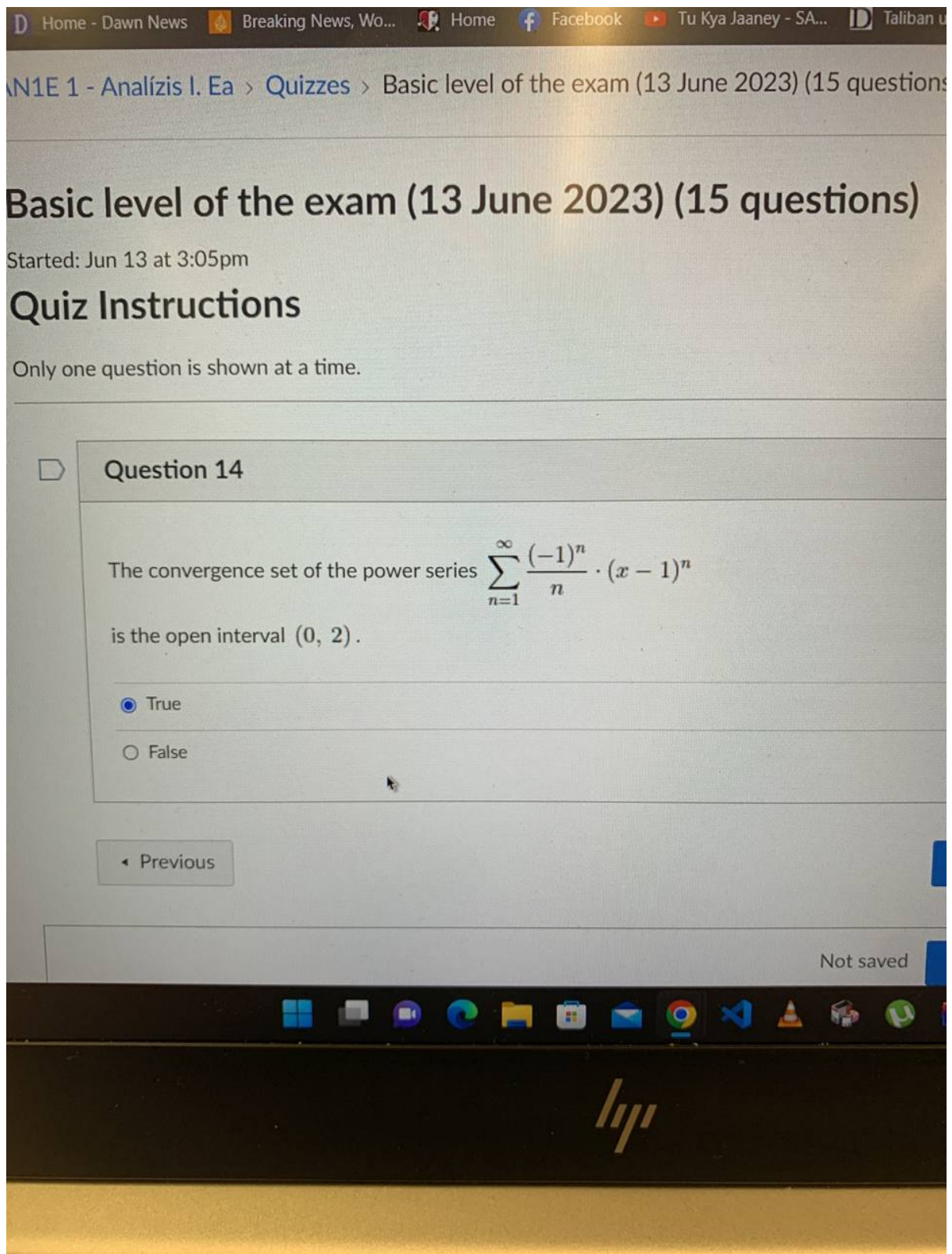
☐ A)

☐ B)

☒ D)

☐ C)

a



false

8. Which of the following power series has the convergence set $(-1, 1]$?

(a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$

(c) $\sum_{n=1}^{\infty} x^n$

(d) $\sum_{n=1}^{\infty} (-1)^n x^n$



Question 1

1 pts

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, and suppose that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and that $\lim_{x \rightarrow -\infty} f(x) = -\infty$. Which of the following statements is true?

A) The function f is monotonically increasing.

B) The equation $f(x) = 0$ has at least one solution in the set of real numbers.

☒ C) There exists an example for f such that the range of f is not the whole set of real numbers (that is: $R_f \neq \mathbb{R}$).

D) The range of f is always $\mathbb{R}$.

Question 1**1 pts**

2. Choose the statement which is **false** :

- (a) If the sequence (a_n) is divergent then (a_n) is not bounded.
- (b) If the limit of a real sequence (a_n) is $+\infty$, then (a_n) is divergent.
- (c) There is no such real sequence (a_n) , which is unbounded below and whose limit is $+\infty$.
- (d) If the real sequence (a_n) is not bounded, then (a_n) is divergent.

☒ A)

☐ B)

☐ D)

☐ C)

20 June 2023) (7 quiz questions

Started: Jun 20 at 4:03pm

Quiz Instructions



Question 5

1 pts

Let $\emptyset \neq A \subseteq \mathbb{R}$.

Which of the following statements is true?



- ☐ If A has maximum and it has finite infimum, then A is bounded set.
- ☐ If A has finite infimum, then it has minimum.
- ☐ If A is bounded, then A has maximum.
- ☐ If A has neither minimum nor maximum, then A is not bounded.

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z Instructions

Question 6

1 pts

4. Choose the statement which is true :

- (a) Let $\emptyset \neq A \subset \mathbb{R}$ and suppose that $\forall \varepsilon > 0 \exists a \in A : a < 2 - \varepsilon$. Then $\sup A = 2$.
- (b) There exists a set $A \subset \mathbb{R}$ which has at least two elements, and for which holds $\inf A = \sup A$.
- (c) Let $\emptyset \neq A \subset \mathbb{R}$ be a bounded set. Then $\sup (A \cup (0, 1)) = \max\{\sup A; 1\}$.
- (d) There is no such nonempty and bounded set $\emptyset \neq A \subset \mathbb{R}$, for which holds $\forall \varepsilon > 0 \forall a \in A : a > \sup A - \varepsilon$.

☐ A)

☐ B)

☒ C)

☐ D)

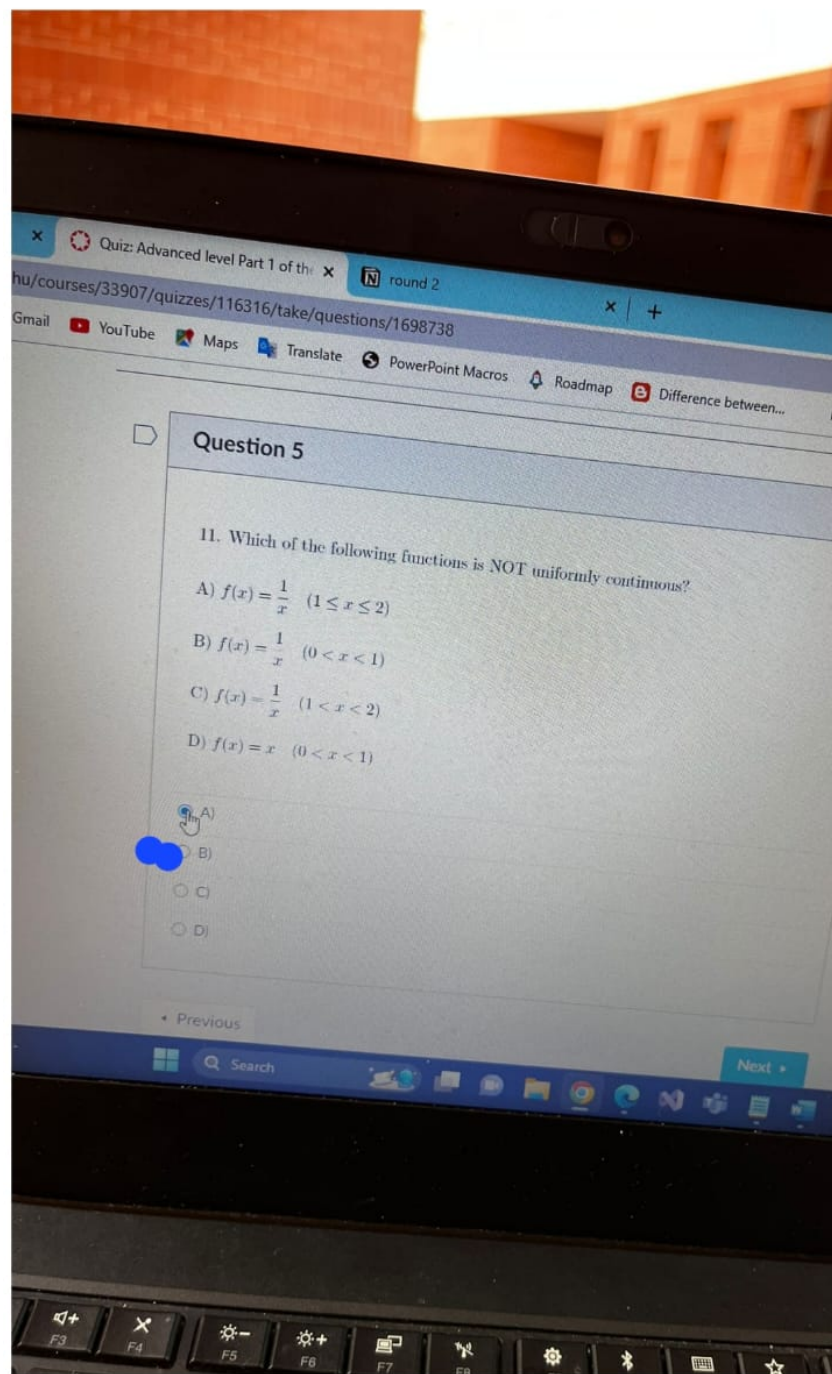
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§ 1 & 2

